Lecture 8 · Uncertainty II: Dominance, States, Ambiguity

MWG Ch. 6.D–6.F · ranking distributions, and losing the probabilities

Advanced Microeconomics 1

Monday 26 October

Today: both goals execute, the ladder runs out, the course closes

1. **Dominance (6.D)**: welfare statements that need *no* recovered preference — rankings every agent in a class signs. The strongest (b)-statements we ever get.
2. **Goal (b), executed (6.E)**: the insurance theorem on your own budget lines — then the Baily–Chetty formula: an optimal social-insurance benefit from two estimable statistics.
3. **Goal (a), executed — then the last rung (6.F)**: your 25 rounds reveal the room's risk attitudes (CFGK); Savage rebuilds probabilities from behavior; your jars show where he can't.

Stochastic dominance

First-order stochastic dominance

Proposition · MWG 6.D.1

F **first-order stochastically dominates** G $\iff$ $F(x) \leq G(x)$ for all x $\iff$ $\int u\, dF \geq \int u\, dG$ for *every* nondecreasing u.

- "F puts more mass on higher outcomes, everywhere." Equivalent construction: F is G plus upward shifts of probability mass.
- Note what FOSD is *not*: it does not say every realization under F beats every realization under G, and it is a **partial** order — most pairs of distributions are simply not ranked. When it does rank, the conclusion is unanimous across all monotone tastes: the strongest kind of welfare statement we ever get.

Second-order stochastic dominance

Proposition · Rothschild–Stiglitz; MWG 6.D.2

For F, G with the *same mean*, the following are equivalent:

1. $\int_{-\infty}^{x} F(t)\,dt \;\le\; \int_{-\infty}^{x} G(t)\,dt$ for all x;
2. $\int u\,dF \ge \int u\,dG$ for every **concave** nondecreasing u: all risk-averse agents prefer F;
3. G is F plus a **mean-preserving spread**: mass moved from the center to the tails.

- The honest warning: variance is not the right ranking. Same mean and higher variance does *not* imply SOSD-dominated — one can build same-mean pairs, unranked by SOSD, where some concave utilities prefer the higher-variance one (EC3 makes you construct such a pair). SOSD demands agreement from *every* concave u; a single summary statistic can't deliver that.

A cameo: inequality measurement is SOSD

Application · Atkinson (JET 1970)

- Read an income distribution as a lottery over "who you are." Comparing societies at equal mean income by *every* concave social welfare function is *exactly* the SOSD comparison — and condition (1) of the last slide is the **Lorenz curve** ordering.
- So "inequality fell, unambiguously" is a mean-preserving-contraction claim; when Lorenz curves cross, no summary index (Gini included) can settle it without taking a stand on curvature.
- One theorem, three literatures: portfolio choice, insurance design, and the measurement of inequality are running the same mathematics.

States and insurance

State-contingent commodities

- Replace abstract lotteries with **states of the world** $s = 1, \ldots, S$ (probabilities π_s for now objective) and **state-contingent claims**: x_s = consumption if state s occurs. A "commodity" is now indexed by the state — Ch. 2's definition, cashing its promise.
- Extended EU: $U(x_1, \ldots, x_S) = \sum_s \pi_s u_s(x_s)$. **State-dependent** u_s matters when the state itself changes the value of money: health, bereavement, weather for a farmer.
- With state-*independent* u and two states, preferences over (x_1, x_2) are an ordinary Ch. 3 consumer — indifference curves in the state-claim plane, with slope $-\pi_1 u'(x_1)/\pi_2 u'(x_2)$, hence slope $-\pi_1/\pi_2$ on the 45° line.

The insurance theorem — and your budget lines

Wealth w, loss D with probability π; insurance at premium rate q per unit of coverage α:

$$\max_{\alpha}\ (1-\pi)\, u(w-q\alpha)+\pi\, u(w-D-q\alpha+\alpha).$$

Proposition · full insurance at fair odds (MWG 6.C.1-adjacent)

If the premium is **actuarially fair** ($q=\pi$) and u is strictly concave, the optimum is $\alpha^*=D$: *full insurance* — equal consumption in both states. If $q>\pi$, coverage is partial.

Proof. FOC at $q=\pi$: $u'(w-\pi\alpha)=u'(w-D+(1-\pi)\alpha)$; strict concavity forces the arguments equal. □

- Geometrically: fair odds make the budget slope $-\pi/(1-\pi)$, the indifference-curve slope *on the 45° line* — a risk-averse consumer rides to certainty.
- **Goal (a), executed on this room:** your Session 1 allocations were exactly this problem — 25 budget lines over two equally likely states (rounds 6, 10 and 13 had *exactly* equal rates). Theory puts every risk-averse one of you *on the 45° line* there: [insert: class scatter at the equal-rate rounds; share within 5 tokens of equal split]

Goal (b), executed: the Baily–Chetty formula

Application · Chetty (JPubE 2006): "A general formula for the optimal level of social insurance"

The insurance theorem scaled to policy. Unemployment benefits *b* funded by taxes; moral hazard prices the coverage. At the optimum, the consumption-smoothing gain equals the incentive cost:

$$\underbrace{r_A \cdot \frac{\Delta c}{c}}_{\text{value of smoothing (Arrow–Pratt)}} \approx \underbrace{\varepsilon_{\text{duration},\,b}}_{\text{moral-hazard elasticity}}$$

- Read the left side: Thursday's currency doing policy work — risk aversion × the consumption drop at job loss. Read the right: a behavioral elasticity, the same object bunching estimated in Chapter 3. Two sufficient statistics, one benefit level.
- This is Hausman's move again, one block later: theory reduces the policy question to estimable objects and outputs a number governments argue about. Full

Subjective probability and ambiguity

Where do the probabilities come from?

- Everything since September assumed probabilities are *given*. For a coin, fine; for "my sector is automated by 2035," there is no objective π.
- **Savage's program** (1954), in one slide: primitives are *acts* (maps from states to outcomes) and preferences over acts. Axioms — centrally the **sure-thing principle**, independence's state-space twin: if two acts agree on some event, the ranking cannot depend on *what* they agree on there.
- Conclusion: preferences satisfying the axioms behave *as if* the agent holds a unique personal probability π and maximizes subjective expected utility. Beliefs are not assumed — they are revealed, exactly as tastes were in Ch. 1. (Anscombe–Aumann 1963 reach the same place faster by allowing objective randomizing devices.)

Ellsberg: the axiom you refused

Two urns, 100 balls each:

- **Known**: 50 red, 50 black.
- **Unknown**: 100 red and black, *mix unknown*.

Bet 1: win if **red**. Bet 2: win if **black**.

Modal pattern: the *known* urn, **both times**.

- But betting known-red reveals your subjective $P(\text{red}_U) < \frac{1}{2}$, and known-black reveals $P(\text{black}_U) < \frac{1}{2}$ — summing to less than one. No probability exists. The sure-thing principle is violated.

- This is not risk aversion — u never appears in the argument. It is **ambiguity aversion**: a distaste for not knowing the odds themselves. Models: maxmin EU over a *set* of priors (Gilboa–Schmeidler 1989); smooth ambiguity (Klibanoff–Marinacci–Mukerji 2005).
- The last two quiz questions in September were exactly these two bets — with the urns played by jars. This room: [insert: share choosing the known jar on both — the Ellsberg pattern].

One more thing you did in September

- Round 24 of the quiz repeated round 4 **word for word**. Round 25 repeated round 12's prices, but *reworded* — abstract accounts instead of the coin.
- Distance between your two answers, same person: [insert: distribution of $d(4, 24)$ vs. $d(12, 25)$; medians]. The first gap is noise; the excess in the second is **framing** — the description of the choice, not the choice, moving behavior.
- Honest caveat (find it before I say it): the two comparisons use *different price vectors*, so baseline noise may differ — the design is suggestive, not clean. What redesign fixes it?
- Why it matters beyond us: Chen et al. (2023) found GPT's rationality score moved with prompt framing. Whatever consistency indices measure, they measure it *jointly with the frame* — in silicon and in this room.

Application · Choi, Fisman, Gale & Kariv (AER 2007) — this block's (a) paper

The parent design of your quiz: portfolio choices on randomly drawn state-contingent budget lines, estimating risk attitudes *jointly* with consistency, subject by subject. Preferences are heterogeneous, mostly well-behaved, and recoverable from the budget lines alone. You have lived inside this paper since week one — it answered Chapter 1's goal (a) at the practical, and Chapter 6's today.

- In passing: Barseghyan et al. (2013) run the same identification on households' real deductible choices — probability distortions, not curvature, do most of the work. Sydnor's puzzle, dissected in the field.

Closing the course

The whole course on one slide: two goals, three blocks

Block	Goal (a): tested	Goal (b): executed	T
Consumer (1–3)	GARP/CCEI — *on you*; S sym. NSD	Hausman: exact CV, EV, DWL	W
Firms (5)	dispersion (Syverson); wedges	Hsieh–Klenow: TFP from reallocation	B
Uncertainty (6)	CFGK rounds — *yours*; Allais	insurance; Baily–Chetty benefit	R

- Three blocks, one loop each time: formalise, buy assumptions, test (a), recover, execute (b), then pay the bill — much of it with data this room generated in twenty minutes in September.
- One craft underneath: pose an optimization, read its value function's curvature, differentiate with an envelope, take the FOC to data. When you meet an empirical claim, ask *which value function was differentiated, and what had to be true for that envelope to hold.*

Prelim survival: the ten cold derivations

1. WARP $\Rightarrow$ compensated law of demand (the four-line expansion).
2. Concavity of $e(p, u)$ in p — and its two siblings, π convex and c concave in w: one trick, three functions.
3. Shephard's lemma and Roy's identity via the envelope theorem.
4. The Slutsky equation from $h(p, u) = x(p, e(p, u))$; the full property list of S.
5. CV and EV as integrals of Hicksian demand; exactness under quasilinearity.
6. Afriat: the inequalities $\Rightarrow$ the min-of-affine rationalizing utility.
7. Hotelling's lemma and the two-line law of supply — plus *why* no compensation is needed.
8. The Cobb–Douglas cost function from scratch; the wedge condition $\text{MRP}_i = (1 + \tau_i)w$ from the distorted FOC.
9. The EU theorem's calibration construction; uniqueness up to positive affine transformations.
10. $\pi \approx \frac{1}{2}\sigma^2 r_A$ by Taylor; SOSD $\iff$ mean-preserving spread $\iff$ all concave agree; full insurance at fair odds.

What comes next, and Thursday

- **Thursday (final session)**: Exercise Class 3 — first hour on Ch. 6 (sheet is out), second hour on past-exam questions across the whole course. Bring the synopsis and the ten derivations.
- Where this hands off:
  - *General equilibrium*: today's supporting-price logic economy-wide; state-contingent commodities become asset markets.
  - *Game theory and IO*: drop price-taking; μ becomes an equilibrium object.
  - *Behavioral and decision theory*: everything our reveals broke, rebuilt one axiom at a time.
- And the closing anecdote this chapter has earned: the author of the 1961 ambiguity paper, having spent a career on decision-making under uncertainty, resolved his own hardest decision in 1971 — Daniel Ellsberg leaked the Pentagon Papers. Uncertainty about the state of the world, professionally, and then literally.

Readings

Thank you — now go differentiate a value function.